

Multiple Choice (10 marks)

1. What endogenous substrate source provides the most energy during moderate to high intensity exercise?
 - (a) Liver glycogen
 - (b) Muscle glycogen
 - (c) Intramuscular lipid
 - (d) Adipose tissue lipid
2. Which of the following foods do not contain free sugars
 - (a) Cows' milk
 - (b) Natural unsweetened orange juice
 - (c) Honey
 - (d) Agave nectar
3. Which of the following vitamins is required for fatty acid β -oxidation?
 - (a) Niacin
 - (b) Riboflavin
 - (c) Thiamin
 - (d) Vitamin C
4. Which of the following statements about glucose metabolism is correct?
 - (a) Glucagon increases the rate of glycolysis
 - (b) Glycolysis requires NADP⁺
 - (c) In glycolysis, glucose is cleaved into two three-carbon products
 - (d) The end-product of glycolysis in red blood cells is pyruvate
5. Which of the following is a function of folate?
 - (a) Synthesis of TMP (thymidine monophosphate)
 - (b) Decarboxylation of amino acids to form amine neurotransmitters
 - (c) Synthesis of methionine from homocysteine
 - (d) Carboxylation of pyruvate to oxaloacetate

True / False (10 marks)

Answer True or False

6. True or False: The 'fear of becoming fat' in patients with anorexia nervosa is best classified as a paranoid delusion affecting their mental state.
7. True or False: A baked apple has a lower glycaemic index than a raw potato.
8. True or False: Older adults usually experience an increase in muscle mass as they age.
9. True or False: Muscle contains less water than fat, which slows down the travel of electrical currents.
10. True or False: Sodium is known to improve hypertension in humans when consumed in high amounts.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the relationship between dietary factors, specifically the consumption of salt and salt-preserved foods, and the increased risk of stomach cancer.
12. Discuss the impact of in utero exposure to the Dutch Hunger Winter on DNA methylation in adult offspring, particularly focusing on the significance of the first trimester of pregnancy.

End of Paper